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Course: EEC 115

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Paper 1 of 1 — EEC 115 ELECTRICAL ENGINEERING SCIENCE

Level ND1 • First Semester • Handout

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PETROLEUM TRAINING INSTITUTE

ELECTRICAL ENGINEERING SCIENCE (EEC 112)

1.1 INTRODUCTION

Given an electrical network, the network analysis involves various methods. The process of finding the network variables namely the voltage and currents in various parts of the circuit is known as network analysis. Before we carry out actual analysis it is very much essential to thoroughly understand the various terms associated with the network. In this chapter we shall begin with the definition and understanding in detail some of the commonly used terms. The basic laws such as Ohm's law, KCL and KVL, those can be used to analyse a given network. Analysis becomes easier if we can simplify the given network. We will be discussing various techniques, which involve combining series and parallel connections of R, L and C elements.

1.2 SYSTEMS OF UNITS

As engineers, we deal with measurable quantities. Our measurement must be communicated in standard language that virtually all professionals can understand irrespective of the country. Such an international measurement language is the International System of Units (SI). In this system, there are six principal units from which the units of all other physical quantities can be derived.

One great advantage of SI unit is that it uses prefixes based on the power of 10 to relate larger and smaller units to the basic unit.

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1.4 ATOM

Definition: An atom is the smallest indivisible particle which all matter is made of up.

1.5 STRUCTURE AND COMPOSITION OF AN ATOM

According to the electron theory of matter, an atom is made up of tiny particles. The central part of the atom is known as the nucleus. The nucleus of the atom consists of particles called protons and neutrons held together in a compact form by a binding energy. The outer portion of the atom is made up of orbiting particles called electrons. Electrons normally move about the centre of an atom in paths which are referred to as shells orbits. Each of these shells can contain only a certain maximum number of electrons. The central part of the atom shows its nucleus. We should note that protons are positively charged particles while electrons are negatively charged particles, since opposite charges attract, the negatively charged electrons are attracted to the positively charged protons in the nucleus. Consequently, this electric force of attraction holds the electrons in their orbit.

The number of orbiting electrons normally equals the number of protons in the nucleus. For

that reason an atom is said to be neutrally charged.

1.6 CONDUCTORS, INSULATORS AND SEMICONDUCTOR

Conductor

In a conductor, electric current can flow freely, in an insulator it cannot. Metals such as copper typify conductors, while most non-metallic solids are said to be good insulators, having extremely high resistance to the flow of charge through them. "Conductor" implies that the outer electrons of the atoms are loosely bound and free to move through the material.

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In copper, the valence electrons are essentially free and strongly repel each other. Any external influence which moves one of them will cause a repulsion of other electrons which propagates, "domino fashion" through the conductor.

Insulator

An Insulator is a material or object which resists the flow of heat (thermal insulators) or electric charge (electrical insulators).

The term insulator has the same meaning as the term dielectric, but the two terms are used in different contexts. The opposite of insulators are conductors and semiconductors, which permit the flow of charge. Semiconductors are strictly speaking also insulators, since they prevent the flow of electric charge at low temperatures, unless doped with atoms that release extra charges to carry the current). However, some materials (such as silicon dioxide) are very nearly perfect electrical insulators, which allow flash memory technology. A much larger class of materials, (for example rubber and many plastics) are "good enough" insulators to be used for home and office wiring (into the hundreds of volts) without noticeable loss of safety or efficiency.

Semiconductor

A semiconductor is a material with an insulator and a conductor.

A semiconductor behaves as an insulator at very low temperature, and has an appreciable conductance at room temperature.

1.7 BASIC CONCEPTS AND DEFINITIONS

CHARGE

The most basic quantity in an electric circuit is the electric charge. We all experience the effect of electric charge when we try to remove our wool sweater and have it stick to our body or walk across a carpet and receive a shock.

Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C). Charge, positive or negative, is denoted by the letter q or Q .

We know from elementary physics that all matter is made of fundamental building blocks known as atoms and that each atom consists of electrons, protons, and neutrons. We also know that the charge 'e' on an electron is negative and equal in magnitude to 1.602×10^{-19} C, while a proton carries a positive charge of the same magnitude as the electron and the neutron has no

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charge. The presence of equal numbers of protons and electrons leaves an atom neutrally charged.

Current

Current can be defined as the motion of charge through a conducting material, measured in Ampere (A). Electric current, is denoted by the letter i or I .

The unit of current is the ampere abbreviated as (A) and corresponds to the quantity of total charge that passes through an arbitrary cross section of a conducting material per unit second.

Mathematically, $I = \frac{Q}{t}$

Where Q is the symbol of charge measured in Coulombs (C), I is the current in amperes (A) and t is the time in second (s).

Two types of currents:

- 1) A direct current (DC) is a current that remains constant with time.
- 2) An alternating current (AC) is a current that varies with time.

Fig. 1.2T wo common types of current: (a) direct current (DC), (b) alternative current (AC)

Example 1.1

Determine the current in a circuit if a charge of 80 coulombs passes a given point in 20 seconds (s).

Example 1.2

How much charge is represented by 4,600 electrons?

Solution:

Each electron has - 1.602×10^{-19} C. Hence 4,600 electrons will have:

$$-1.602 \times 10^{-19} \times 4600 = -7.369 \times 10^{-16} \text{ C}$$

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1.8 ANALOGY BETWEEN CURRENT FLOW AND WATER FLOW

The three main concepts in electricity are voltage, current and resistance. Voltage or potential difference can be thought of as the driving force (although it is not really a "force") behind the electric current. In the water analogy, in figure (3.2), you can think of voltage as the pressure difference created by a pump that causes the water to flow through the pipe in the water system. Also you can think of the electrons flow in the electric circuit as equivalent to the flow of the water in the pipes that caused by the pump.

Another important concept in electricity is resistance. It's Property of a material that opposes the flow of electrons through it. There is always a resistance flow between two points in a pipe. This is analogous to the resistance (R) in an electric circuit

1.9 VOLTAGE (or) POTENTIAL DIFFERENCE

To move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external electromotive force (emf), typically represented by the battery in Fig. 1.3. This emf is also known as voltage or potential difference. The voltage ab between two points a and b in an electric circuit is the energy (or work) needed to move a unit charge from a to b .

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Fig. 1.3(a) Electric Current in a conductor, (b) Polarity of voltage ab

Voltage (or potential difference) is the energy required to move charge from one point to the other, measured in volts (V). Voltage is denoted by the letter v or V .

Mathematically, $V = \frac{W}{Q}$

where w is energy in joules (J) and q is charge in coulombs (C). The voltage ab or simply V is measured in volts (V).

1 volt = 1 joule/coulomb = 1 newton-meter/coulomb

Example 1.5

An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 KJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

Solution:

Total charge

$$Q = I \times t = 2 \times 10 = 20 \text{ C}$$

$$2.3 \times 10^3$$

$$V = \frac{W}{Q} = \frac{2300}{20}$$

$$115 \text{ V}$$

1.10 POWER

Power is the time rate of expending or absorbing energy, measured in watts (W). Power, is denoted by the letter p or P .

Mathematically, $P = \frac{W}{t}$

Where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s).

From voltage and current equations, it follows that

$$P = \frac{W}{t} = \frac{QV}{t} = \frac{Q}{t} \times V = I \times V$$

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Thus, if the magnitude of current I and voltage are given, then power can be evaluated as the product of the two quantities and is measured in watts (W).

Sign of power:

Plus sign: Power is absorbed by the element. (Resistor, Inductor)

Minus sign: Power is supplied by the element. (Battery, Generator)

Passive sign convention:

If the current enters through the positive polarity of the voltage, $p = +vi$

If the current enters through the negative polarity of the voltage, $p = -vi$

Polarities for Power using passive sign convention

(a) Absorbing Power (b) Supplying Power

1.11 ENERGY

Energy is the capacity to do work, and is measured in joules (J).

The energy absorbed or supplied by an element from time 0 to t is given by,

$$W = \int_0^t p \, dt$$

The electric power utility companies measure energy in watt-hours (WH) or Kilo watt-hours (KWH)

$$1 \text{ WH} = 3600 \text{ J}$$

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Example 1.6

A source e.m.f. of 5 V supplies a current of 3A for 10 minutes. How much energy is provided in this time?

Solution:

$$W = 5 \times 3 \times 10 \times 60 = 27000 \text{ J}$$

Example 1.7

An electric heater consumes 1.8MJ when connected to a 250 V supply for 30 minutes. Find the power rating of the heater and the current taken from the supply.

Solution:

$$P = (1.8 \times 10^6) / (30 \times 60) = 1000$$

Power rating of heater = 1kW

Thus

$$I = 1000/250 = 4$$

Hence the current taken from the supply is 4A.

1.12 OHM'S LAW

Georg Simon Ohm (1787–1854), a German physicist, is credited with finding the relationship between current and voltage for a resistor. This relationship is known as Ohm's law.

Ohm's law states that at constant temperature, the voltage (V) across a conducting material is directly proportional to the current (I) flowing through the material.

Mathematically,

$$V \propto I$$

$$V = IR$$

Where the constant of proportionality R is called the resistance of the material. The V-I relation for resistor according to Ohm's law is depicted in Fig.1.6

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Remember the following three rules:

Ohm's Law can only give the correct answer when the correct values are used.

Current is always expressed in Amperes or Amps (A)

Voltage is always expressed in Volts (V)

Resistance is always expressed in Ohms (Ω)

Below is an Easy way to remember which formula to use

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Limitations of Ohm's Law:

1. Ohm's law is not applicable to non-linear elements like diode, transistor etc.
2. Ohm's law is not applicable for non-metallic conductors like silicon carbide.

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1.13 CIRCUIT ELEMENTS

An element is the basic building block of a circuit. An electric circuit is simply an interconnection of the elements. Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit.

There are 2 types of elements found in electrical circuits.

a) Active elements (Energy sources): The elements which are capable of generating or delivering the energy are called active elements.

E.g., Generators, Batteries

b) Passive element (Loads): The elements which are capable of receiving the energy are called passive elements.

E.g., Resistors, Capacitors and Inductors

Active Elements (Energy Sources)

The energy sources which are having the capacity of generating the energy are called active elements. The most important active elements are voltage or current sources that generally deliver power/energy to the circuit connected to them.

There are two kinds of sources

- a) Independent sources
- b) Dependent sources

Independent sources:

An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

Ideal Independent Voltage Source:

An ideal independent voltage source is an active element that gives a constant voltage across its terminals irrespective of the current drawn through its terminals. In other words, an ideal independent voltage source delivers to the circuit whatever current is necessary to maintain its terminal voltage. The symbol of idea independent voltage source and its V-I characteristics are shown in Fig. 1.7

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Practical Independent Voltage Source:

Practically, every voltage source has some series resistance across its terminals known as internal resistance, and is represented by R_{se} . For ideal voltage source $R_{se} = 0$. But in practical voltage source R_{se} is not zero but may have small value. Because of this R_{se} voltage across the terminals decreases with increase in current as shown in Fig. 1.8

Terminal voltage of practical voltage source is given by

$$V_L = V_S - I_L R_{se}$$

Practical Independent Voltage Source

Ideal Independent Current Source:

An ideal independent Current source is an active element that gives a constant current through its terminals irrespective of the voltage appearing across its terminals. That is, the current source delivers to the circuit whatever voltage is necessary to maintain the designated current. The symbol of idea independent current source and its V-I characteristics are shown in Fig. 1.9

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Figure Ideal Independent Current Source

Practical Independent Current Source:

Practically, every current source has some parallel/shunt resistance across its terminals known as internal resistance, and is represented by R_{sh} . For ideal current source $R_{sh} = \infty$ (infinity). But in practical voltage source R_{sh} is not infinity but may have a large value. Because of this R_{sh} current through the terminals slightly decreases with increase in voltage across its terminals as shown in Fig. 1.10.

Terminal current of practical current source is given by

$$I_L = I_s - I_{sh}$$

Practical Independent Current Source

Passive Elements (Loads)

Passive elements are those elements which are capable of receiving the energy. Some passive

elements like inductors and capacitors are capable of storing a finite amount of energy, and return it later to an external element. More specifically, a passive element is defined as one that cannot supply average power that is greater than zero over an infinite time interval. Resistors, capacitors, Inductors fall in this category.

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1.14 RESISTOR

Materials in general have a characteristic behavior of opposing the flow of electric charge. This physical property, or ability to resist the flow of current, is known as resistance and is represented by the symbol R . The Resistance is measured in ohms (Ω). The circuit element used to model the current resisting behavior of a material is called the resistor.

Fig. 1.13 (a) Typical Resistor, (b) Circuit Symbol for Resistor

The resistance of a resistor depends on the material of which the conductor is made and geometrical shape of the conductor. The resistance of a conductor is proportional to the its length and inversely proportional to its cross-sectional area (A). Therefore the resistance of a conductor can be written as,

$$R = \rho l / A$$

The proportionality constant ρ is called the resistivity of the conductor and its value depends on the material of which the conductor is made.

The inverse of the resistance is called the conductance and inverse of resistivity is called specific conductance or conductivity. The symbol used to represent the conductance is G and conductivity is σ .

Thus conductivity $\sigma = 1/\rho$ and its units are Siemens per meter

Example 1.8

In the circuit shown in Fig. below, calculate the current i , the conductance G , the power p and energy lost in the resistor W in 2 hours.

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PROBLEM: Calculate the resistance of 100 m length of a wire having a uniform cross-sectional area of 0.1 mm^2 if the wire is made of manganin having a resistivity of $50 \times 10^{-8} \Omega\text{-m}$. If the

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wire is drawn out to three times its original length, by how many times would you expect its resistance to be increased ?

Effect of Temperature on Resistance

The effect of rise in temperature is:

(i) to increase the resistance of pure metals. The increase is large and fairly regular for normal ranges of temperature. The temperature/resistance graph is a straight line

As would be presently clarified, metals have a positive temperature co-efficient of resistance.

(ii) to increase the resistance of alloys, though in their case, the increase is relatively small and irregular. For some high-resistance alloys like Eureka (60% Cu and 40% Ni) and manganin, the increase in resistance is (or can be made) negligible over a considerable range of temperature.

(iii) to decrease the resistance of electrolytes, insulators (such as paper, rubber, glass, mica etc.) and partial conductors such as carbon. Hence, insulators are said to possess a negative temperature-coefficient of resistance.

Temperature Coefficient of Resistance

Let a metallic conductor having a resistance of R_0 at 0°C be heated of $t^\circ\text{C}$ and let its resistance at this temperature be R_t . Then, considering normal ranges of temperature, it is found that the increase in resistance $\Delta R = R_t - R_0$ depends

- i. directly on its initial resistance
- ii. directly on the rise in temperature
- iii. on the nature of the material of the conductor.

$$R_t - R_0 \propto R_0 \times t \text{ or } R_t - R_0 = \alpha R_0 t$$

where α (alpha) is a constant and is known as the temperature coefficient of resistance of the conductor

Rearranging, we get

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If $R_0 = 1 \Omega$, $t = 1^\circ\text{C}$, then $\alpha = \Delta R = R_t - R_0$ Hence, the temperature-coefficient of a material may be defined as: the increase in resistance per ohm original resistance per $^\circ\text{C}$ rise in temperature.

we find that $R_t = R_0 (1 + \alpha t)$

Value of α at Different Temperatures

So far we did not make any distinction between values of α at different temperatures. But it is found that value of α itself is not constant but depends on the initial temperature on which the increment in resistance is based. When the increment is based on the resistance measured at 0°C , then α has the value of α_0 . At any other initial temperature $t^\circ\text{C}$, value of α is α_t and so on. It should be remembered that, for any conductor, α_0 has the maximum value.

In case R_0 is not given, the relation between the known resistance R_1 at $t_1^\circ\text{C}$ and the unknown resistance R_2 at $t_2^\circ\text{C}$ can be found as follows :

1.15 INDUCTOR

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Fig. 1.14 (a) Typical Inductor, (b) Circuit symbol of Inductor

A wire of certain length, when twisted into a coil becomes a basic inductor. The symbol for inductor is shown in Fig.1.14 (b). If current is made to pass through an inductor, an electromagnetic field is formed. A change in the magnitude of the current changes the electromagnetic field. Increase in current expands the fields, and decrease in current reduces it. Therefore, a change in current produces change in the electromagnetic field, which induces a voltage across the coil according to Faraday's law of electromagnetic induction. i.e., the voltage across the inductor is directly proportional to the time rate of change of current.

Mathematically,

$$V \propto \frac{dI}{dt}$$

$$V \propto L \frac{dI}{dt}$$

$$V = L \frac{dI}{dt}$$

$$V dt = L dI$$

$$\int_0^t V dt = L \int_0^I dI$$

$$V t = L I$$

Where L is the constant of proportionality called the inductance of an inductor. The unit of inductance is Henry (H).we can rewrite the above equation as

$$V = L \frac{dI}{dt}$$

$$V dt = L dI$$

$$\int_0^t V dt = L \int_0^I dI$$

Integrating both sides from time 0 to t, we get

$$V dt = L dI$$

$$\int_0^t V dt = L \int_0^I dI$$

$$\int_0^t V dt = L \int_0^I dI$$

$$V t = L I$$

$$V dt = L dI$$

$$\int_0^t V dt = L \int_0^I dI + I(0)$$

$$V t = L I$$

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From the above equation we note that the current in an inductor is dependent upon the integral of the voltage across its terminal and the initial current in the coil.

The power absorbed by the inductor is

$$P = VI$$

$$P = L I \frac{dI}{dt}$$

$$P = L I \frac{dI}{dt}$$

The energy stored by the inductor is

$$W = \int P dt$$

$$W = \int_0^t P dt$$

$$W = \int_0^t P dt$$

$\frac{dI}{dt}$

$\frac{dI}{dt} = 0$

$V = \int_0^L \frac{dI}{dt} dx =$

0

From the above discussion, we can conclude the following.

1. The induced voltage across an inductor is zero if the current through it is constant. That means an inductor acts as short circuit to DC.
2. A small change in current within zero time through an inductor gives an infinite voltage across the inductor, which is physically impossible. In a fixed inductor the current cannot change abruptly i.e., the inductor opposes the sudden changes in currents.
3. The inductor can store finite amount of energy. Even if the voltage across the inductor is zero
4. A pure inductor never dissipates energy, only stores it. That is why it is also called a non-dissipative passive element. However, physical inductors dissipate power due to internal resistance.

Example

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1.16 CAPACITOR

Fig. 1.15 (a) Typical Capacitor, (b) Capacitor connected to a voltage source, (c) Circuit Symbol of capacitor

Any two conducting surfaces separated by an insulating medium exhibit the property of a capacitor. The conducting surfaces are called electrodes, and the insulating medium is called dielectric. A capacitor stores energy in the form of an electric field that is established by the opposite charges on the two electrodes. The electric field is represented by lines of force between the positive and negative charges, and is concentrated within the dielectric.

When a voltage source v is connected to the capacitor, as in Fig 1.15 (b), the source deposits a positive charge q on one plate and a negative charge $-q$ on the other. The capacitor is said to store the electric charge. The amount of charge stored, represented by q , is directly proportional to the applied voltage v so that $q = Cv$

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Where C , the constant of proportionality is known as the capacitance of the capacitor. The unit of capacitance is the farad (F).

The current flowing through the capacitor is given by

$i = \frac{dq}{dt}$

$= C \frac{dv}{dt}$

$i = C \frac{dv}{dt}$

$$i = C \frac{dv}{dt}$$

$$i = C \frac{dv}{dt}$$

$$i = C \frac{dv}{dt}$$

We can rewrite the above equation as

$$1$$

$$i = C \frac{dv}{dt}$$

$$i = C \frac{dv}{dt}$$

Integrating both sides from time 0 to t, we get

$$1$$

$$i = \int \frac{dv}{dt} + v(0)$$

$$i = 0$$

The power absorbed by the capacitor is

$$P = i v$$

$$P = i v = i v$$

$$P = i v$$

The energy stored by the capacitor is

$$W = \int P dt$$

$$W = \int P dt$$

$$W = 0$$

$$W = 0$$

$$W = \frac{1}{2} C v^2$$

$$W = \int P dt =$$

$$0 \frac{1}{2} C v^2$$

From the above discussion we can conclude the following,

1. The current in a capacitor is zero if the voltage across it is constant; that means, the capacitor acts as an open circuit to DC.
2. A small change in voltage across a capacitance within zero time gives an infinite current through the capacitor, which is physically impossible. In a fixed capacitance the voltage cannot change abruptly. i.e., A capacitor will oppose the sudden changes in voltages.
3. The capacitor can store a finite amount of energy, even if the current through it is zero.

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4. A pure capacitor never dissipates energy, but only stores it; that is why it is called non-dissipative passive element. However, physical capacitors dissipate power due to internal resistance.

1.17 NETWORK/CIRCUIT TERMINOLOGY

In the following section various definitions and terminologies frequently used in electrical circuit analysis are outlined.

- ❖ Network Elements: The individual components such as a resistor, inductor, capacitor, diode, voltage source, current source etc. that are used in circuit are known as network elements.
- ❖ Network: The interconnection of network elements is called a network.
- ❖ Circuit: A network with at least one closed path is called a circuit. So, all the circuits are networks but all networks are not circuits.
- ❖ Branch: A branch is an element of a network having only two terminals.
- ❖ Node: A node is the point of connection between two or more branches. It is usually indicated by a dot in a circuit.
- ❖ Loop: A loop is any closed path in a circuit. A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.
- ❖ Mesh or Independent Loop: Mesh is a loop which does not contain any other loops in it.

1.18 SERIES AND PARALLEL CIRCUITS

Series circuits

A series circuit is formed when any numbers of resistors are connected end-to-start so that there is only one path for current to flow. The resistors can be actual resistors or other devices that have resistance like lamps.

Figure (6.1) shows two resistors connected in series (end -to -start) between points A and B. There is only way for current to get from point A to point B. a series circuit provides only one path for current between two points in a circuit so that the current in the same through R1 and R2.

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Parallel Circuits

A parallel circuit is formed when any number of resistors are connected starts-to-starts and ends-to-ends so that there is different paths for current to flow. The resistors can be actual resistors or other devices that have resistance like lamps Figure shows two resistors connected in parallel (start-to-start and end-to-end) between points A and B. There is two ways for current to get from point A to point B. A parallel circuit provides more than one path for current between two points in a circuit so that the current flow in two paths current flow in R1 is called I1 and current flow in R2 is called I2.

Resistors in Series

$$R = R_1 + R_2 + R_3 \text{ ohm}$$

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where R is the total circuit resistance. From this result it may be seen that when resistors are connected in series the total resistance is found simply by adding together the resistor values.

Resistors in Parallel

1 1 1 1

= + +

□ □1 □2 □3

Potential Divider

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Current Divider

Example

For the circuit shown in Fig. calculate (a) the current drawn from the supply, (b) the current through the 6 Ω resistor, and (c) the power dissipated by the 5.6 Ω resistor.

For the circuit of Fig. calculate (a) the current drawn from the source, (b) the p.d. across each resistor, (c) the current through each resistor, and (d) the power dissipated by the 5 Ω resistor.

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2.1 KIRCHHOFF'S LAWS

The most common and useful set of laws for solving electric circuits are the Kirchhoff's voltage and current laws. Several other useful relationships can be derived based on these laws. These laws are formally known as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL).

Kirchhoff's Current Law (KCL)

This is also called as Kirchhoff's first law or Kirchhoff's nodal law. Kirchhoff's first law is based on the law of conservation of charge, which requires that the algebraic sum of charges within a system cannot change.

Statement: Algebraic sum of the currents meeting at any junction or node is zero. The term 'algebraic' means the value of the quantity along with its sign, positive or negative.

Mathematically, KCL implies that

Alternate Statement: Sum of the currents flowing towards a junction is equal to the sum of the currents flowing away from the junction.

Applying Kirchhoff's current law to the junction A

$I_1 + I_2 - I_3 + I_4 - I_5 = 0$ (algebraic sum is zero)

The above equation can be modified as $I_1 + I_2 + I_4 = I_3 + I_5$ (sum of currents towards the junction = sum of currents flowing away from the junction).

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Kirchhoff's Voltage Law (KVL)

This is also called as Kirchhoff's second law or Kirchhoff's loop or mesh law. Kirchhoff's second law is based on the principle of conservation of energy.

Statement: Algebraic sum of all the voltages around a closed path or closed loop at any instant is zero. Algebraic sum of the voltages means the magnitude and direction of the voltages; care should be taken in assigning proper signs or polarities for voltages in different sections of the circuit.

Mathematically, KVL implies that

Where N is the number of voltages in the loop (or the number of branches in the loop) and is the nth voltage in a loop.

For example consider the circuit shown in Fig.

For loop1 it is considered around clockwise

$$+ E1 - V1 + V3 - E3 = 0$$

$$+ E1 - I1 R1 + I3 R3 - E3 = 0$$

$$E1 - E3 = I1 R1 - I3 R3$$

For loop2 it is considered anticlockwise

$$+ E2 + V2 + V3 - E3 = 0$$

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$$+ E2 + I2 R2 + I3 R3 - E3 = 0$$

$$E2 - E3 = - I2 R2 - I3 R3$$

Two equations are obtained following Kirchhoff's voltage law. The third equation can be written based on Kirchhoff's current law as

$$I1 - I2 + I3 = 0$$

With the three equations, one can solve for the three currents I1, I2, and I3. If the results obtained for I1, I2, and I3 are all positive, then the assumed direction of the currents are said to be along the actual directions. A negative result for one or more currents will indicate that the assumed direction of the respective current is opposite to the actual direction.

Examples

1. For the circuit of Fig. 2.19 determine the value and direction of the current in each branch, and the p.d. across the 10 Ω resistor.

2. Use KVL to determine the unknown voltages V1 and V4 in the circuit of Figure 2.21.

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3. Calculate the current supplied by two batteries in the circuit given below

2.1 SUPERPOSITION PRINCIPLE

The superposition principle is a method that allows us to determine the current through or the voltage across any resistor or branch in a network. The advantage of using this approach instead of Kirchhoff's laws is that it is not necessary to use determinant or matrix algebra to analyze a given circuit. The theorem states the following:

In a circuit containing several sources of e.m.f., the resultant current in any branch is the algebraic sum of the currents in that branch that would be produced by each e.m.f, acting alone, all other sources of e.m.f being replaced meanwhile by their respective internal resistances.

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In order to apply the superposition principle it is necessary to remove all sources other than the one being examined. In order to "Zero" a voltage source, we replace it with a short circuit, since the voltage across a short circuit is zero volts. A current source is zeroed by replacing it with an open circuit, since the current through an open circuit is zero amps.

Example

In the circuit below determine all the branch currents using the superposition principles.

2.2 ELECTRIC FIELDS

Electrostatics is the branch of science, which deals with the phenomena associated with electricity at rest.

Under normal circumstances an atom is electrically neutral which means that the aggregate of positive charge of protons is exactly equal to the aggregate of negative charge of electrons.

But if somehow some electrons are removed from atoms of body, then it is left with a preponderance of positive charge and the body is said to be positively charged. If on the other hand, some electrons are added to the atoms of a body, then negative charge out-balance the positive charge and the body is said to be negatively charged.

Therefore, the deficiency of electrons of the atoms in a body makes it to be positively charged while the excess of electrons of the atoms in a body makes it negatively charged, so the charge is known as the total deficiency or excess of electrons in a body and the unit is in Coulomb (abbreviated C).

Coulombs Laws of Electrostatics

1. First Law: like charges of electricity repels each other, whereas unlike charges attract each other.

2. Second Law: the force exerted between two point charges is;

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(i) directly proportional to the product of their strength.

(ii) inversely proportional to the square of the distance between them

This is known as the coulombs laws of electrostatics and can be expressed mathematically as;

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$\frac{1}{4}$

where k is the constant of proportionality, $k = \frac{1}{4\pi\epsilon}$ where ϵ = permittivity of the medium.

$\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$

$\epsilon =$

$4\pi\epsilon_0 \frac{q_1 q_2}{r^2}$

$\frac{1}{4\pi\epsilon_0}$

$\epsilon = 9 \times 10^9$

$\frac{1}{r^2}$

Examples

- i. Calculate the electrostatic force of repulsion between two α -particles when at a distance of 10^{-13} m from each other. Charge of an α -particles is 3.2×10^{-12} C.
- ii. The small identical conducting spheres have charges of 2.0×10^{-9} C and -0.5×10^{-9} respectively. When they are placed 4 cm apart, what is the force between them? If they are brought into contact and then separated by 4 cm, what is the force between them?
- iii. Calculate the distance of separation between two electrons for which the electric force between them is 0.1N. Charge of electron is 1.6×10^{-19} C.

2.3 DEFINITION OF TERMS USED IN ELECTROSTATICS

Electric Field

The region in which the stress exists or in which electric forces act, is called an electric field or electrostatic field. The stress is represented by imaginary lines of forces

Electric field strength

When a P.d of V volts is applied between two metals plates or electrodes, which are d meters apart an electric field is established between the plates. The arrangement in fig 14.1 is referred to as parallel-plate capacitor. The medium separating the plates is an insulator, and it is known

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as the dielectric. A feature of the capacitor is that they are capable of storing energy, and this energy is stored in the dielectric material of the capacitor.

The electric field strength is the total voltage acting on a field per length of the line (field).

The electric field strength, symbol E , in the dielectric is given by the equation

$$E = V/d, (V/m)$$

Example

Two conducting plates are arranged parallel to each other and spaced 1.5mm apart. If the plates are charged until the potential difference between them is 150V, what is the electric field strength between the plates?

Solution

$E =$ 150

$$\phi = 100 \times 10^{-3}$$

$$\phi = 1.5 \times 10^{-3}$$

Electric flux density

One line of electric flux is assumed to emanate from a positive charge of one coulomb, hence Q lines of flux emanate from a charge of Q coulomb. Therefore, Electric flux = Q, it is equal to the stored charge. If the flux passes through a dielectric of area A (the area being measured perpendicular to the direction of the flux), then the electric flux density is the amount of electric flux passing through a small area A at right angle to the flux per unit area, it has symbol D.

Mathematically

$$D = \phi/A = \Psi/A \text{ (C/m}^2\text{)}$$

Example

Two plates, each of area 5cm², are placed parallel to each other and very close together, and a charge of 15 X 10⁻⁸C is stored on the plates. Calculate the electric flux density in the space between the plates.

Permittivity

The permittivity, symbol ϵ , of a dielectric material is a measure of the ability of the material to allow an electric flux to be established in it. It has the dimensions of farad per meters' (f/m). For a given value, a material with a high value of permittivity produces a greater electrostatic flux than does a material with a lower value of permittivity for a simple parallel plate capacitor.

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Permittivity, $\epsilon = \text{Electric flux density} / \text{Electric field strength} = D/E$

$$\epsilon = \phi/E$$

Permittivity is also given by

$$\epsilon = \epsilon_0 \epsilon_r$$

Relative permittivity

Relative permittivity (ϵ_r) is the ratio of the absolute permittivity to the permittivity of free space.

Note that ϵ_r has no dimension because ϵ_r and ϵ_0 have the same units

2.5 CAPACITANCE

A capacitor is an arrangement which has the ability to store electricity as excess of positive charges on one plate and deficiency of it on the other. Therefore the property of a capacitor to store electric charge when its plates are at different potentials is reflected to as its capacitance.

The unit of capacitance is termed the farad (symbol f) and farad can be defined as the capacitance of a capacitor between the plates of which there appears a difference of potential of 1 volt when it is charged by 1 coulomb of electricity.

Capacitance (farads) = Charge (coulombs)/Applied voltage (volts)

$$C = Q/V$$

Capacitance of Parallel Plate Capacitors in terms of Area, the distance between Plates and permittivity of the dielectric is given as

$$C = \frac{\epsilon A}{d}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

For parallel capacitors with n numbers of plates

$$C = (n - 1) \frac{\epsilon A}{d}$$

$$C = \frac{\epsilon A}{d}$$

$$C = \frac{\epsilon A}{d}$$

where n is the number of plates.

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2.6 CAPACITOR CONNECTIONS

Parallel connection

Capacitors in a parallel configuration each have the same potential difference (voltage). The reason for putting capacitors in parallel is to increase the total amount of charged stored. In other words, increasing the capacitance we also increase the amount of energy that can be stored.

The total capacitance (C_{eq}) is given by

$$C_{eq} = C_1 + C_2 + \dots + C_n$$

Series connection

The current through capacitors in series stays the same, but the voltage across each capacitor can be different. The sum of the potential differences (voltage) is equal to the total voltage. The reason for putting capacitors in series we get less capacitance and less charge storage than with either alone

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Capacitors of 3uF, 6 uF and 12 uF are connected in series across a 400 V supply. Determine the p.d. across each capacitor.

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For the circuit shown in the Fig., determine (a) the charge drawn from the supply, (b) the charge on the 8 uF capacitor, (c) the p.d. across the 4 uF capacitor, and (d) the p.d. across the 3 uF capacitor.

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2.7 ENERGY STORED IN A CAPACITOR

When a capacitor is connected to a voltage source of V volts, we have seen that it will charge up until the p.d. between the plates is also V volts. If the capacitor is now disconnected from the supply, the charge and p.d. between its plates will be retained.

Energy stored,

$$W = \frac{1}{2} QV$$

Example

A $3\mu\text{F}$ capacitor is charged from a 250V d.c. supply. Calculate the charge and energy stored. The charged capacitor is now removed from the supply and connected across an uncharged $6\mu\text{F}$ capacitor. Calculate the p.d. between the plates and the energy now stored by the combination.

Solution

$$C_1 = 3\mu\text{F}; C_2 = 6\mu\text{F}; V = 250\text{V}$$

$$Q = C_1 V = 3 \times 10^{-6} \times 250 = 0.75\text{mC}$$

1 1

$$Q = C_1 V = 3 \times 10^{-6} \times 250 = 0.75\text{mC}$$

2 2

When the two capacitors are connected in parallel the $3\mu\text{F}$ will share its charge with $6\mu\text{F}$ capacitor. Thus the total charge in the system will remain unchanged, but the total capacitance will now be different:

$$C_{\text{total}} = C_1 + C_2 = 3\mu\text{F} + 6\mu\text{F} = 9\mu\text{F}$$

$$Q = 0.75 \times 10^{-3}$$

$$V = \frac{Q}{C} = \frac{0.75 \times 10^{-3}}{9 \times 10^{-6}} = 83.33\text{V}$$

$$C = 9 \times 10^{-6}$$

1 1

$$W = \frac{1}{2} QV = \frac{1}{2} \times 0.75 \times 10^{-3} \times 83.33 = 31.25\text{mJ}$$

2 2

$$W = 31.25\text{mJ}$$

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3.1 MAGNETIC MATERIALS

All materials may be broadly classified as being in one of two groups. They may be magnetic or non-magnetic, depending upon the degree to which they exhibit magnetic effects. The vast majority of materials fall into the latter group, which may be further classified into diamagnetic and paramagnetic materials. The magnetic properties of these materials are very slight, and extremely difficult even to detect. Thus, for practical purposes, we can say that they are totally

non-magnetic.

The magnetic materials (based on iron, cobalt and ferrites) are the ferromagnetic materials, all of which exhibit very strong magnetic effects. It is with these materials that we will be principally concerned.

3.2 MAGNETIC FIELDS

Magnetic fields are produced by permanent magnets and by electric current flowing through a conductor. Like the electric field, a magnetic field may be considered as being the medium by which forces are transmitted. In this case, the forces between magnetised materials.

A magnetic field is also represented by lines of force or magnetic flux.

These are attributed with certain characteristics, listed below:

- i. They always form complete closed loops. Unlike lines of electric flux, which radiate from and terminate at the charged surfaces, lines of magnetic flux also exist all the way through the magnet.
- ii. They behave as if they are elastic. That is, when distorted they try to return to their natural shape and spacing.
- iii. In the space surrounding a magnet, the lines of force radiate from the north (N) pole to the south (S) pole.
- iv. They never intersect (cross).
- v. Like poles repel and unlike poles attract each other

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3.3 MAGNETIC FLUX AND FLUX DENSITY

The magnetic flux is what causes the observable magnetic effects such as attraction, repulsion etc. The unit of magnetic flux is the weber (Wb).

The number of webers of flux per square metre of cross-section of the field is defined as the magnetic flux density (B), which is measured in tesla (T).

Worked Examples

3.4 MAGNETOMOTIVE FORCE (MMF)

In an electric circuit, any current that flows is due to the existence of an emf. Similarly, in a magnetic circuit, the magnetic flux is due to the existence of an mmf. The concept of an mmf for permanent magnets is a difficult one. Fortunately, it is simple when we consider the flux being produced by current flowing through a coil. This is the case for most practical magnetic circuits.

□□□, □ = □□ □□□□□ □□□□

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Worked example

3.5 MAGNETIC FIELD STRENGTH

magnetic field strength is defined as the mmf per metre length of the magnetic circuit. The quantity symbol for magnetic field strength is H , the unit of measurement being ampere turn/metre.

$\square \square$ turn

$\square =$ = ampere

$\square \square$ metre

Where l is the mean or average length of the magnetic circuit.

3.6 PERMEABILITY

The 'willingness' of A material to allow for the establishment of a magnetic field is the permeability of the material. If the magnetic field exists in a vacuum, then the ratio of the flux density to the magnetic field strength is a constant, called the permeability of free space.

$\square$

$$\square_0 = h\square\square\square\square/\square\square\square\square$$

$\square$

Absolute Permeability

The absolute permeability of a material is the ratio of the flux density to magnetic field strength, for a given mmf.

$\square$

$$\square = \square h\square\square\square\square/\square\square\square\square$$

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Relative permeability

For measuring relative permeability, vacuum or free space is chosen as the reference medium. It is allotted an absolute permeability of $\mu_0 = 4\pi \times 10^{-7}$ henry/metre. Obviously, relative permeability of vacuum with reference to itself is unity. Hence, for free space, absolute permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m relative permeability $\mu_r = 1$. Now, take any medium other than vacuum. If its relative permeability, as compared to vacuum is μ_r , then its absolute permeability is $\mu = \mu_0 \mu_r$ H/m.

Worked example

3.7 B-H CURVES: (MAGNETIZATION CURVES)

For ferromagnetic materials, μ is not constant but varies with flux density and there is no easy way to compute it. In reality, however, it isn't μ that you are interested in: What you really want to know is, given B , what is H , and vice versa. A set of curves, called B-H or magnetization curves, provides this information. (These curves are obtained experimentally and are available in handbooks. A separate curve is required for each material.) Figure shows typical curves for cast iron, cast steel, and sheet steel B-H curve (magnetization curves)

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3.8 AMPERE'S CIRCUITAL LAW:

If we apply the analogy to Kirchhoff's voltage law (for magnetic circuits).

States that the algebraic sum of the rises and drops of the mmf around a closed loop of a magnetic circuit is equal to zero; that is, the sum of the rises in mmf equals the sum of the drops in mmf around a closed loop. Equation is referred to as Ampère's circuital law. When it is applied to magnetic circuits, sources of mmf are expressed by the equation. The summation is algebraic and terms are additive or subtractive, depending on the direction of flux and how the coils are wound.

Consider the magnetic circuit appearing in Figure constructed of three different ferromagnetic materials. Applying Ampère's circuital law, we have

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Which states that the applied mmf NI is equal to the sum of the drops HL around the loop. The path to use for the HL terms is the mean (average) path.

You now have two magnetic circuit models (Figure below). While the reluctance model (a) is not very useful for solving problems, it helps relate magnetic circuit problems to familiar electrical circuit concepts. The Ampere's law model, on the other hand, permits us to solve practical problems.

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3.9 MAGNETIC CIRCUIT

It may be defined as the route or path which is followed by magnetic flux. The laws of magnetic circuit are quite similar to (but not the same as) those of the electric circuit. Consider a solenoid or a toroidal iron ring having a magnetic path of l metre, area of cross section A m² and a coil of N turns carrying I amperes wound anywhere on it as in the Fig. Field strength inside the solenoid is

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μ_0

μ_r

$\mu = \mu_0 \mu_r$

H

$B = \mu H$

Φ

$\Phi = \mu_0 \mu_r N I = \mu N I$

Φ^2

$\mu_0 \mu_r N^2 I^2$

The total flux produce $\Phi = \mu \times I = \mu N I$

Φ

Φ

$$\Phi = \frac{NI}{\frac{l}{\mu}}$$

$\frac{l}{\mu}$

$$\frac{NI}{\frac{l}{\mu}}$$

The numerator 'NI' which produces magnetization in the magnetic circuit is known as magnetomotive force (mmf). Obviously, its unit is ampere-turn (AT). It is analogous to e.m.f. in an electric circuit.

$\frac{l}{\mu}$

The denominator $\frac{l}{\mu}$ is called the reluctance of the circuit and is analogous to resistance in electric circuits.

$\Phi = \frac{NI}{\frac{l}{\mu}}$

Therefore flux = m.m.f / reluctance or $\Phi = \frac{NI}{\frac{l}{\mu}}$

Sometimes, the above equation is called the "Ohm's Law of Magnetic Circuit" because it resembles a similar expression in electric circuits i.e.

$$\text{current} = \frac{\text{e.m.f.}}{\text{resistance}} \text{ or } I = \frac{V}{R}$$

3.10 COMPARISON BETWEEN MAGNETIC AND ELECTRIC CIRCUITS

Similarities

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Differences

1. Strictly speaking, flux does not actually 'flow' in the sense in which an electric current flow.
2. If temperature is kept constant, then resistance of an electric circuit is constant and is independent of the current strength (or current density). On the other hand, the reluctance of a magnetic circuit does depend on flux (and hence flux density) established in it. It is so because μ (which equals the slope of B/H curve) is not constant even for a given material as it depends on the flux density B. Value of μ is large for low value of B and vice-versa. Hence, reluctance is small ($S = \frac{l}{\mu A}$) for small values of B and large for large values of B.
3. Flow of current in an electric circuit involves continuous expenditure of energy but in a magnetic circuit, energy is needed only creating the flux initially but not for maintaining it.

3.11 DEFINITIONS CONCERNING MAGNETIC CIRCUIT

1. Magnetomotive force (mmf.). It drives or tends to drive flux through a magnetic circuit and corresponds to electromotive force (e.m.f.) in an electric circuit. M.M.F. is equal to the work done in joules in carrying a unit magnetic pole once through the entire magnetic circuit. It is measured in ampere-turns.

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2. Ampere-turns (AT). It is the unit of magnetomotive force (mmf.) and is given by the

product of number of turns of a magnetic circuit and the current in amperes in those turns.

3. Reluctance. It is the name given to that property of a material which opposes the creation of magnetic flux in it. It, in fact, measures the opposition offered to the passage of magnetic flux through a material and is analogous to resistance in an electric circuit even in form. Its unit is AT/Wb.

4. Permeance. It is reciprocal of reluctance and implies the ease or readiness with which magnetic flux is developed. It is analogous to conductance in electric circuits. It is measured in terms of Wb/AT or henry.

5. Reluctivity. It is specific reluctance and corresponds to resistivity which is 'specific resistance'.

Example

A steel toroid of the dimensions csa 4.5cm^2 and mean radius of 30cm shown in is wound with a 500 turns coil of wire. What value of current needs to be passed through this coil in order to produce a flux of $250\text{ }\mu\text{Wb}$ in the toroid, if under these conditions the relative permeability of the toroid is 300.

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Example 6: A solenoid with a core of csa of 15 cm^2 and relative permeability 65, produces a flux of $200\text{ }\mu\text{Wb}$. If the core material is changed to one of relative permeability 800 what will be the new flux and flux density? Ans: $B = 1.641\text{T}$ and $\Phi = 2.462\text{ mWb}$

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3.12 COMPOSITE SERIES MAGNETIC CIRCUIT

Figure 2

In Fig. 2 is shown a composite series magnetic circuit consisting of three different magnetic materials of different permeabilities and lengths and one air gap ($\mu_r = 1$). Each path will have its own reluctance. The total reluctance is the sum of individual reluctances as they are joined in series.

□

□□□□□ □□□□□□□□□□ = Σ

□0 □□ □

□1 □2 □3 □□

= + + +

□0 □□ □1 □0 □□ □2 □0 □□ □3 □0 □□

□. □. □

□□□□ $\Phi =$

□

□0 □□ □

How to Find Ampere-turns

$$\text{Magnetic field strength } H = \frac{B}{\mu} \times \frac{1}{\mu_r}$$

Hence, following procedure should be adopted for calculating the total ampere turns of a composite magnetic path.

- (i) Find H for each portion of the composite circuit. For air, $H = B/\mu_0$, otherwise $H = B/\mu_0\mu_r$.
- (ii) Find ampere-turns for each path separately by using the relation $AT = H \times l$.
- (iii) Add up these ampere-turns to get the total ampere-turns for the entire circuit.

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3.13 PARALLEL MAGNETIC CIRCUITS

Figure 3

Fig. 3 (a) shown a parallel magnetic circuit consisting of two parallel magnetic paths ACB and ADB acted upon by the same m.m.f. Each magnetic path has an average length of $2(l_1 + l_2)$. The flux produced by the coil wound on the central core is divided equally at point A between the two outer parallel paths. The reluctance offered by the two parallel paths is = half the reluctance of each path.

Fig. 3 (b) shows the equivalent electrical circuit where resistance offered to the voltage source is $= R/R = R/2$

3.14 SERIES-PARALLEL MAGNETIC CIRCUITS

Figure 4

Such a circuit is shown in Fig. 4 (a). It shows two parallel magnetic circuits ACB and ACD connected across the common magnetic path AB which contains an air-gap of length l_g . As usual, the flux Φ in the common core is divided equally at point A between the two parallel paths which have equal reluctance. The reluctance of the path AB consists of

- (i) air gap reluctance and

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- (ii) the reluctance of the central core which comparatively negligible.

Hence, the reluctance of the central core AB equals only the air-gap reluctance across which are connected two equal parallel reluctances. Hence, the m.m.f. required for this circuit would be the sum of

- (i) that required for the air-gap and
- (ii) that required for either of two paths (not both).

The equivalent electrical circuit is shown in Fig. 4 (b) where the total resistance offered to

the voltage source is $= R_1 + R/R = R_1 + R/2$.

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3.15 ELECTROMAGNETISM

Most applications of magnetism involve magnetic effects due to electric currents. The current, I , creates a magnetic field that is concentric about the conductor, uniform along its length, and whose strength is directly proportional to I . Note the direction of the field. It may be remembered with the aid of the right-hand rule, imagine placing your right hand around the conductor with your thumb pointing in the direction of current. Your fingers then point in the direction of the field. If you reverse the direction of the current, the direction of the field reverses.

If the conductor is wound into a coil (Provided no ferromagnetic material is present), the fields of its individual turns combine, producing a resultant field as in Figure. The direction of the coil flux can also be remembered by means of a simple rule: curl the fingers of your right hand around the coil in the direction of the current and your thumb will point in the direction of the field. If the direction of the current is reversed, the field also reverses. Provided no ferromagnetic material is present, the strength of the coil's field is directly proportional to its current.

If the coil is wound on a ferromagnetic core as in Figure below (transformers are built this way), almost all flux is confined to the core, although a small amount (called stray or leakage flux) passes through the surrounding air. However, now that ferromagnetic material is present, the core flux is no longer proportional to current.

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3.2 FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

It is mainly due to the pioneering work of Michael Faraday, in the nineteenth century, that the modern technological world exists as we know it. Without the development of the generation of electrical power, such advances would have been impossible. Thus, although the concepts involved with electromagnetic induction are very simple, they have far-reaching influence.

Faraday's law is best considered in two interrelated parts:

- 1 The value of emf induced in a circuit or conductor is directly proportional to the rate of change of magnetic flux linking with it.
- 2 The polarity of such an emf, induced by an increasing flux, is opposite to that induced by a decreasing flux.

Figure 1

$-\frac{d\Phi}{dt}$

$\mathcal{E} = -\frac{d\Phi}{dt} \quad \text{--- (1)}$

$\mathcal{E}$

3.3 LENZ'S LAW

This law states that the polarity of an induced emf is always such that it opposes the change which produced it. This is similar to the statement in mechanics that for every force there is an opposite reaction.

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3.5 FLEMING'S RIGHTHAND RULE

This is a convenient means of determining the polarity of an induced emf in a conductor. Also, provided that the conductor forms part of a complete circuit, it will indicate the direction of the resulting current flow.

The first finger, the second finger and the thumb of the righthand are held out mutually at right angles to each other (like the three edges of a cube as shown in Fig 2a). The First finger indicates the direction of the Flux, the thuMb the direction of Motion of the conductor relative to the flux, and the sECond finger indicates the polarity of the induced Emf, and direction of Current flow. This process is illustrated in Fig. 2b, which shows the cross-section of a conductor being moved vertically upwards at a constant velocity through the magnetic field.

(a) (b)

Figure 2

Example 1: The flux linking a 100 turn coil changes from 5mWb to 10mWb in a time of 5ms. Calculate the average emf induced in the coil.

Example 2: A 250 turn coil is linked by a magnetic flux that varies as follows: an increase from zero to 20mWb in a time of 0.05s; constant at this value for 0.02s; followed by a decrease to 4mWb in a time of 0.01s. Assuming that these changes are uniform, draw a sketch graph (i.e. not to an accurate scale) of the variation of the flux and the corresponding emf induced in the coil, showing all principal values.

3.6 EMF INDUCED IN A SINGLE STRAIGHT CONDUCTOR

Consider a conductor moving at a constant velocity v metre per second at right angles to a magnetic field having the dimensions shown in Fig. 3. The direction of the induced emf may

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be obtained using Fleming's righthand rule, and is shown in the diagram. Equation 1 is applicable, and in this case, the value for N is 1.

Figure 3

Thus

$$d\Phi$$

$$\square =$$

$$\square \square$$

But flux is constant

$$\Phi$$

$$\mathcal{E} =$$

$$\mathcal{E}$$

$$\Phi = \mathcal{B}A$$

$$\mathcal{B}A$$

$$\mathcal{E} =$$

$$\mathcal{E}$$

The cross-sectional area of the field $\mathcal{A} = \mathcal{L} \times \mathcal{W}$

$$\mathcal{B}l$$

$$\mathcal{E} =$$

$$\mathcal{E}$$

$$\mathcal{E}$$

$$\mathcal{E} = \mathcal{B}l v \sin \theta \quad \text{--- (2)}$$

$$\mathcal{E}$$

The above equation is only true for the case when the conductor is moving at right angles to the magnetic field. If the conductor moves through the field at some angle less than 90° , then the 'cutting' action between the conductor and the flux is reduced. This results in a consequent reduction in the induced emf. Thus, if the conductor is moved horizontally through the field, the 'cutting' action is zero, and so no emf is induced. To be more precise, we can say that only the component of the velocity at 90° to the flux is responsible for the induced emf. In general therefore, the induced emf is given by

$$\mathcal{E} = \mathcal{B}l v \sin \theta$$

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where $v \sin \theta$ is the component of velocity at 90° to the field, as illustrated in Fig. 4.

Figure 4

Example 3: A conductor is moved at a velocity of 5m/s at an angle of 60° to a uniform magnetic field of 1.6mWb. The field is produced by a pair of pole pieces, the faces of which measure 10cm by 4cm. If the conductor length is parallel to the longer side of the field, calculate the emf induced.

wave, i.e. an alternating voltage.

3.7 FORCE ON A CURRENT-CARRYING CONDUCTOR

Figure 7(a) shows the field patterns produced by two pole pieces, and the current flowing through the conductor. Since the lines of flux obey the rule that they will not intersect, then the flux pattern from the poles will be distorted as illustrated in Fig. 7(b). Also, since the lines of flux tend to act as if elastic, then they will try to straighten themselves. This results in a force being exerted on the conductor, in the direction shown.

$$\mathcal{F} = \mathcal{B}l i \sin \theta \quad \text{--- (3)}$$

Any conductor extending beyond the main field does not contribute to the force exerted.

Example 4: A pair of pole pieces 5cm by 3cm produce a flux of 2.5mWb. A conductor is placed

in this field with its length parallel to the longer dimension of the poles. When a current is passed through the conductor, a force of 1.25N is exerted on it. Determine the value of the current.

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If the conductor was placed at 45° to the field, what then would be the force exerted?

3.8 FORCE BETWEEN PARALLEL CONDUCTORS

When two parallel conductors are both carrying current their magnetic fields will interact to produce a force of attraction or repulsion between them. This is illustrated in Fig. 9

Figure 9

In order to determine the value of such a force, consider first a single conductor carrying a current of I ampere. The magnetic field produced at some distance d from its centre is shown in Fig. 10.

Figure 10

□□ □□□□

In general, $\square = \square \square \square \square \square \square \square \square \square \square$

But in this case, $N = 1$ and $l = 2\pi d$, so

□□

□=

$2\square\square$

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Now, flux density $\square = \square 0 \square \square \square$ tesla, and as the field exists in air, then $\mu_r = 1$. Thus, the flux density at distance d from the centre is given by

□□ □

□=

$2\square\square$

Consider now two conductors Y and Z carrying currents I_1 and I_2 respectively, at a distance of d metres between their centres as in Fig 11

Figure 11

we can say that the flux density acting on Z due to current I_1 flowing in Y is:

□□ □1

□=

$2\square\square$

And the force exerted on Z = $B I_2 l$ newton, or $B I_2$ newton per meter length of Z

Hence force/meter length acting on Z

□□ □1 □2

$$2 \times 10^{-7} \times 1 \times 2$$

$$4 \times 10^{-7} \times 1 \times 2$$

$$2 \times 10^{-7} \times 1 \times 2$$

$$2 \times 10^{-7} \times 1 \times 2$$

□

Now, the current I_2 flowing in Z also produces a magnetic field which will exert a force on Y.

Using the same reasoning as above, it can be shown that:

force/metre length acting on Y

$$2 \times 10^{-7} \times 1 \times 2$$

□

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If $I_1 = I_2 = 1\text{A}$ and $d = 1\text{m}$

Force exerted on each conductor $= 2 \times 10^{-7} \text{ N/m}$

This value of force forms the basis for the definition of the ampere, namely: that current, which when maintained in each of two infinitely long parallel conductors situated in vacuum, and separated one metre between centres, produces a force of $2 \times 10^{-7} \text{ N/m}$ per metre length on each conductor.

Example 5: Two long parallel conductors are spaced 35mm between centres. Calculate the force exerted between them when the currents carried are 50 A and 40 A respectively.

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3.9 SELF AND MUTUAL INDUCTANCE

The effects of self and mutual inductance can be demonstrated by another simple experiment. Consider two coils, as shown in Figure 1. Coil 1 can be connected to a battery via a switch. Coil 2 is placed close to coil 1, but is not electrically connected to it. Coil 2 has a galvo connected to its terminals.

Figure 1

When the switch is closed, the current in coil 1 will rapidly increase from zero to some steady value. Hence, the flux produced by coil 1 will also increase from zero to a steady value. This changing flux links with the turns of coil 2, and therefore induces an emf into it. This will be indicated by a momentary deflection of the galvo pointer.

Similarly, when the switch is subsequently opened, the flux produced by coil 1 will collapse to zero. The galvo will again indicate that a momentary emf is induced in coil 2, but of the opposite polarity to the first case. Thus, an emf has been induced into coil 2, by a changing current (and flux) in coil 1. This is known as a mutually induced emf.

If the changing flux can link with coil 2, then it must also link with the turns of coil 1. Thus, there must also be a momentary emf induced in this coil. This is known as a self-induced emf.

Any induced emf obeys Lenz's law. This self-induced emf must therefore be of the opposite polarity to the battery emf. For this reason, it is also referred to as a back emf. Unfortunately, it is extremely difficult to demonstrate the existence of this back emf. If a voltmeter was connected across coil 1, it would merely indicate the terminal voltage of the battery.

3.10 SELF-INDUCTANCE

Self-inductance is that property of a circuit or component which causes a self-induced emf to be produced, when the current through it changes. The unit of self-inductance is the henry, which is defined as follows:

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A circuit has a self-inductance of one henry (1 H) if an emf of one volt is induced in it, when the circuit current changes at the rate of one ampere per second (1 A/s).

The quantity symbol for self-inductance is L . From the above definition, we can state the following equation

$$-e = L \frac{di}{dt}$$

$$e = h \frac{d\phi}{dt}$$

$$\frac{d\phi}{dt}$$

$$\frac{d\phi}{dt}$$

Exercise: A coil has a self-inductance of 0.25 H. Calculate the value of emf induced, if the current through it changes from 100 mA to 350 mA, in a time of 25ms.

Exercise: Calculate the inductance of a circuit in which an emf of 30 V is induced, when the circuit current changes at the rate of 200 A/s.

3.11 SELF-INDUCTANCE AND FLUX LINKAGES

Consider a coil of N turns, carrying a current of I amp. Let us assume that this current produces a flux of ϕ weber. If the current now changes at a uniform rate of di/dt ampere per second, it will cause a corresponding change of flux of $d\phi/dt$ weber per second. Let us also assume that the coil has a self-inductance of L henry. The self-induced emf may be determined from

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

However, the induced emf is basically due to the rate of change of flux linkages

$$e = -N \frac{d\phi}{dt}$$

$$e = -N \frac{d\phi}{dt}$$

$$e = -N \frac{d\phi}{dt}$$

Since both equations represent the same induced emf, then

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

A coil which is designed to have a specific value of self-inductance is known as an inductor

If the current I through an N turn coil produces a flux of ϕ weber, then its self-inductance is given by the equation

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

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Exercise: A coil of 150 turns carries a current of 10 A. This current produces a magnetic flux of 0.01 Wb. Calculate (a) the inductance of the coil, and (b) the emf induced when the current is uniformly reversed in a time of 0.1 s.

Exercise: A current of 8 A, when flowing through a 3000turn coil, produces a flux of 4 mWb. If the current is reduced to 2 A in a time of 100 ms, calculate the emf thus induced in the coil. Assume that the flux is directly proportional to the current.

3.12 FACTORS AFFECTING INDUCTANCE

Consider a coil of N turns wound on to a non-magnetic core, of uniform cross-sectional area A metre² and mean length L metre. The coil carries a current of 1 amp, which produces a flux of ϕ weber. We know that the inductance will be

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}, \text{ But } \phi = BA$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\text{Therefore } \Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}, \text{ But } \phi = BA$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}, \text{ But } \phi = BA$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

$$\Phi = \frac{1}{\mu_0 \mu_r} \frac{N^2 I^2}{l}$$

Φ

$\Phi_{12} =$

$\Phi_{21} =$

Φ_{11}

$\Phi_{22} =$

Φ

Example: A 600 turn coil is wound on to a non-magnetic core of effective length 45 mm and csa 4 cm². (a) Calculate the inductance (b) The number of turns is increased to 900. Calculate the inductance value now produced. (c) The core of the 900 turns coil is now replaced by an iron core having a relative permeability of 75, and of the same dimensions as the original. Calculate the inductance in this case

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3.13 MUTUAL INDUCTANCE

When a changing current in one circuit induces an emf in another separate circuit, then the two circuits are said to possess mutual inductance. The unit of mutual inductance is the henry, and is defined as follows. Two circuits have a mutual inductance of one henry, if the emf induced in one circuit is one volt, when the current in the other is changing at the rate of one ampere per second. The quantity symbol for mutual inductance is M, and expressing the above definition as an equation we have

Induced emf in coil 2

$\mathcal{E}_2 =$

Induced emf in coil 1

$-\frac{d\Phi_{21}}{dt}$

$\mathcal{E}_1 =$

$-\frac{d\Phi_{12}}{dt}$

Φ_{12}

$-\frac{d\Phi_{12}}{dt}$

$\mathcal{E}_2 = M \frac{dI_1}{dt}$

Φ_{12}

This emf may also be expressed in terms of the flux linking coil 2. If all of the flux from coil 1 links with coil 2, then we have what is called 100% flux linkage. In practice, it is more usual for only a proportion of the flux from coil 1 to link with coil 2. Thus the flux linkage is usually less than 100%. This is indicated by a factor, known as the coupling factor, k. 100% coupling is indicated by $k = 1$. If there is no flux linkage with coil 2, then k will have a value of zero. So if zero emf is induced in coil 2, the mutual inductance will also be zero. Thus, the possible values for the coupling factor k, lie between zero and 1. Expressed mathematically, this is written as

$0 \leq k \leq 1$

Consider two coils possessing mutual inductance, and with a coupling factor less than 1. Let a change of current di_1/dt amp/s in coil 1 produce a change of flux $d\phi_1/dt$ weber/s. The proportion of this flux change linking coil 2 will be $d\phi_2/dt$ weber/s. If the number of turns on coil 2 is N_2 , then

$$-N_2 \frac{d\phi_2}{dt}$$

$$\phi_2 = \frac{M}{N_2} \frac{d\phi_1}{dt}$$

$$\frac{d\phi_2}{dt} = \frac{M}{N_2} \frac{d^2\phi_1}{dt^2}$$

$$\frac{d^2\phi_1}{dt^2} = \frac{N_2}{M} \frac{d\phi_2}{dt}$$

$$\frac{d\phi_2}{dt} = \frac{M}{N_2} \frac{d^2\phi_1}{dt^2}$$

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$$\phi_2 = \frac{M}{N_2} \frac{d\phi_1}{dt}$$

$$M = h \frac{d\phi_1}{dt}$$

$$\frac{d\phi_1}{dt} = \frac{M}{h}$$

As with self-inductance for a single coil, mutual inductance is a property of a pair of coils. They therefore retain this property, regardless of whether or not an emf is induced. Hence, equation may be modified to

$$\phi_2 = \frac{M}{N_2} \frac{d\phi_1}{dt}$$

$$\frac{d\phi_2}{dt} = \frac{M}{N_2} \frac{d^2\phi_1}{dt^2}$$

$$\frac{d^2\phi_1}{dt^2} = \frac{N_2}{M} \frac{d\phi_2}{dt}$$

Example: Two coils, A and B, have 2000 turns and 1500 turns respectively. A current of 0.5 A, flowing in A, produces a flux of 60Wb. The flux linking with B is 83% of this value.

Determine (a) the self-inductance of coil A, and (b) the mutual inductance of the two coils.

RELATIONSHIP BETWEEN SELF- AND MUTUAL-INDUCTANCE

Consider two coils of N_1 and N_2 turns respectively, wound on to a common non-magnetic core. If the reluctance of the core is S ampere turns/weber, and the coupling coefficient is unity, then

$$\phi_{12} = \phi_{22}$$

$$\phi_1 = \frac{N_1 I_1}{S} \quad \phi_2 = \frac{N_2 I_2}{S}$$

$$\phi_1 = \phi_2$$

Therefore

$$\phi_{12} = \phi_{22}$$

$$\phi_1 \phi_2 = \frac{N_1 N_2 I_1 I_2}{S^2} \quad (1)$$

$$\phi_2 = \frac{N_2 I_2}{S}$$

$$\phi_2 = \frac{N_2 I_2}{S}$$

$$M = h \frac{d\phi_1}{dt}, \quad \text{then } \frac{d\phi_1}{dt} = \frac{M}{h}$$

$$\phi_1 = \frac{M}{h}$$

$$\phi_1 \phi_2 = \frac{M}{h} \phi_2$$

$$\phi_2 = \frac{M}{h}$$

$$\phi_1 = \frac{M}{h}$$

$$1 \Phi$$

$$\Phi_1 \Phi_1$$

$$\Phi_1 \Phi_2$$

$$\Phi =$$

$$\Phi$$

$$\Phi_1 \Phi_2$$

$$\Phi_2 = \dots \dots \dots (2)$$

$$\Phi_2$$

Comparing equation 1 and 2

$$\Phi_2 = \Phi_1 \Phi_2$$

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Therefore

$$\Phi = \sqrt{\Phi_1 \Phi_2}$$

The above equation is correct only provided that there is 100% coupling between the coils; i.e. $k = 1$. If k is less than 1, then the general form of the equation, shown below

$$\Phi = k\sqrt{\Phi_1 \Phi_2}$$

Example: A 400 turn coil is wound onto a cast steel toroid having an effective length of 25 cm and csa 4.5 cm². If the steel has a relative permeability of 180 under the operating conditions, calculate the self-inductance of the coil.

3.14 INDUCTANCES IN SERIES

Let the two coils be so joined in series that their fluxes (or mmfs) are additive i.e., in the same direction.

Figure 2

Let M = coefficient of mutual inductance

L_1 = coefficient of self-inductance of 1st coil

L_2 = coefficient of self-inductance of 2nd coil.

$$\Phi = \Phi_1 + \Phi_2 + 2\Phi$$

When the coils are so joined that their fluxes are in opposite directions

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$$\Phi = \Phi_1 + \Phi_2 - 2\Phi$$

Example: Two coils with a coefficient of coupling of 0.5 between them, are connected in series so as to magnetise (a) in the same direction (b) in the opposite direction. The corresponding values of total inductances are for (a) 1.9 H and for (b) 0.7 H. Find the self-inductances of the two coils and the mutual inductance between them.

3.15 INDUCTANCE IN PARALLEL

Figure 4

In Fig. 4, two inductances of values L_1 and L_2 henry are connected in parallel. Let the coefficient of mutual inductance between the two be M . Let i be the main supply current and i_1 and i_2 be the branch currents. Obviously, $i = i_1 + i_2$

$$V_1 = V_2 = V$$

$$V = -L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} \quad (3)$$

$$V = -L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

Equation 3 is when mutual field assists the separate fields.

For when the two fields oppose each other

$$V_1 = V_2 = V$$

$$V =$$

$$V = -L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

Example: Two coils of inductances 4 and 6 henry are connected in parallel. If their mutual inductance is 3 henry, calculate the equivalent inductance of the combination if (i) mutual inductance assists the self-inductance (ii) mutual inductance opposes the self-inductance

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A.C. FUNDAMENTALS

This chapter deals with the concepts, terms and definitions associated with alternating quantities. The term alternating quantities refers to any quantity (current, voltage, flux, etc.), whose polarity is reversed alternately with time. For convenience, they are commonly referred to as a.c. quantities.

On completion of this chapter you should be able to:

- Explain the method of producing an a.c. waveform.
- Define all of the terms relevant to a.c. waveforms.
- Obtain values for an a.c., both from graphical information and when expressed in mathematical form.
- Understand and use the concept of phase angle.
- Use both graphical and phasor techniques to determine the sum of alternating quantities.

4.1 INTRODUCTION

Alternating voltage may be generated by rotating a coil in a magnetic field, as shown in Fig. 1 (a) or by rotating a magnetic field within a stationary coil, as shown in Fig. 1 (b).

Figure 1

The value of the voltage generated depends, in each case, upon the number of turns in the coil, strength of the field and the speed at which the coil or magnetic field rotates. Alternating voltage may be generated in either of the two ways shown above, but rotating-field method is the one which is mostly used in practice.

4.2 EQUATIONS OF THE ALTERNATING VOLTAGES AND CURRENTS

Consider a rectangular coil, having N turns and rotating in a uniform magnetic field, with an angular velocity of ω radian/second, as shown in Fig. 2. Let time be measured from the X-axis. Maximum flux Φ_m is linked with the coil, when its plane coincides with the X-axis. In

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time t seconds, this coil rotates through an angle $\theta = \omega t$. In this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\Phi = \Phi_m \cos \omega t$. Hence, flux linkages of the coil at any time are $N \Phi = N \Phi_m \cos \omega t$. According to Faraday's Laws of Electromagnetic Induction, the e.m.f. induced in the coil is given by the rate of change of flux-linkages of the coil. Hence, the value of the induced e.m.f. at this instant (i.e. when $\theta = \omega t$) or the instantaneous value of the induced e.m.f. is

Figure 2

Similarly, the equation of induced alternating current is $i = I_m \sin \omega t$...(iv) provided the coil circuit has been closed through a resistive load. Since $\omega = 2\pi f$, where f is the frequency of rotation of the coil, the above equations of the voltage and current can be written as

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Figure 3

4.3 SIMPLE WAVEFORMS

The shape of the curve obtained by plotting the instantaneous values of voltage or current as the ordinate against time as abscissa is called its waveform or wave-shape.

Figure 4

An alternating voltage or current may not always take the form of a systematical or smooth wave such as that shown in Fig. 3. Thus, Fig. 4 also represents alternating waves. But while it is scarcely possible for the manufacturers to produce sine-wave generators or alternators, yet sine wave is the ideal form sought by the designers and is the accepted standard. The waves deviating from the standard sine wave are termed as distorted waves. In general, however, an alternating current or voltage is one the circuit direction of which reverses at regularly recurring intervals.

Time Period: The time taken by an alternating quantity to complete one cycle is called its time period T . For example, a 50-Hz alternating current has a time period of $1/50$ second.

Frequency: The number of cycles/second is called the frequency of the alternating quantity. Its unit is hertz (Hz).

$f = PN/120$ where N = revolutions in r.p.m. and P = number of poles

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Amplitude: The maximum value, positive or negative, of an alternating quantity is known as its amplitude.

Phase: By phase of an alternating current is meant the fraction of the time period of that alternating current which has elapsed since the current last passed through the zero position of reference.

Root-Mean-Square (R.M.S.) Value

The r.m.s. value of an alternating current is given by that steady (d.c.) current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time. It is also known as the effective or virtual value of the alternating current, the former term being used more extensively.

The standard form of a sinusoidal alternating current is $i = I_m \sin \omega t = I_m \sin \theta$. The mean of the squares of the instantaneous values of current over one complete cycle is (even the value over half a cycle will do).

Hence, we find that for a symmetrical sinusoidal current

$$\frac{1}{T} \int_0^T i^2 dt = 0.707 \times I_m \quad \text{or} \quad I_{\text{R.M.S.}} = 0.707 \times I_m$$

The r.m.s. value of an alternating current is of considerable importance in practice, because the ammeters and voltmeters record the r.m.s. value of alternating current and voltage respectively.

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In electrical engineering work, unless indicated otherwise, the values of the given current and voltage are always the r.m.s. values

Average Value

The average value I_{av} of an alternating current is expressed by that steady current which transfers across any circuit the same charge as is transferred by that alternating current during the same time. In the case of a symmetrical alternating current (i.e. one whose two half-cycles are exactly similar, whether sinusoidal or non-sinusoidal), the average value over a complete cycle is zero. Hence, in their case, the average value is obtained by adding or integrating the instantaneous values of current over one half-cycle only. But in the case of an unsymmetrical alternating current (like half-wave rectified current) the average value must always be taken over the whole cycle.

For

one half cycle

$$Q = \int_0^T i dt = \int_0^{\pi} I_m \sin \theta d\theta$$

$$Q = I_m [-\cos \theta]_0^{\pi} = I_m [1 - (-1)] = 2I_m$$

$$I_{\text{av}} = \frac{Q}{T} = \frac{2I_m}{2\pi} = \frac{I_m}{\pi}$$

$$I_{\text{av}} = \frac{I_m}{\pi} = 0.637 \times I_m$$

$$I_{\text{av}} = 0.637 \times I_m$$

$$I_{\text{av}} = 0.637 \times I_m$$

Note. R.M.S. value is always greater than average value except in the case of a rectangular wave when both are equal.

Form Factor

It is defined as the ratio, $K_f = \text{r.m.s. value} / \text{average value}$

0.701□□

$$\square\square = = 1.11$$

0.637□□

Crest or Peak or Amplitude Factor

It is defined as the ratio $K_a = \text{maximum value} / \text{rms value}$

11

$$\square\square = \sqrt{2} = 1.414$$

$$\frac{1}{\sqrt{2}}$$

Example 1: An alternating current of frequency 60 Hz has a maximum value of 120 A. Write down the

equation for its instantaneous value. Reckoning time from the instant the current is zero and is becoming

positive, find (a) the instantaneous value after $1/360$ second and (b) the time taken to reach 96 A for the

first time.

Solution.

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The instantaneous current equation is $i = 120 \sin 2 \pi \times 10^3 t = 120 \sin 120 \pi t$

Now when $t = 1/360$ second, then

(a)

1

$$y = 120 \sin(120 \times \pi \times t) \dots \text{amplitude} \text{ and frequency}$$

360

1

$$= 120 \sin (120 \times 180 \times) \dots$$

360

$$= 120 \sin 60^\circ = 103.9 \text{ N}$$

(b)

$$96 = 120 \times \frac{1}{2} \times 180 \times 60 \times \frac{1}{2} \times \dots \times \frac{1}{2} \times 180 \times 60 \times \frac{1}{2}$$

96

$$\sin(360 \times 60 \times \square) = 0.8$$

120

$$360 \times 60 \times \pi = 111-1.08 = 53^\circ \text{ (approx)}$$

$$\therefore \theta = \theta/2\pi = 53/360 \times 60 = 0.00245 \text{ radian}.$$

Example 2: An alternating current varying sinusoidally with a frequency of 50 Hz has an RMS value of 20 A. Write down the equation for the instantaneous value and find this value (a) 0.0025 second (b) 0.0125 second after passing through a positive maximum value. At what time, measured from a positive maximum value, will the instantaneous current be 14.14 A?

Example 3: A rectangular coil, measuring 25 cm by 20 cm, has 80 turns. The coil is rotated, about an axis parallel with its longer sides, in a magnetic field of density 75 mT. If the speed of rotation is 3000 rev/min, calculate, from first principles, (a) the amplitude, r.m.s. and average values of the emf, (b) the frequency and period of the generated waveform, (c) the instantaneous value, 2ms after it is zero.

4.4 PHASE DIFFERENCE

When two alternating quantities of the same frequency have different zero points, they are said to have a phase difference. The angle between the zero points is the angle of phase difference.

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In Phase

Two waveforms are said to be in phase, when the phase difference between them is zero. That is the zero points of both the waveforms are same. The waveform, phasor and equation representation of two sinusoidal quantities which are in phase is as shown. The figure shows that the voltage and current are in phase.

$$v = V_m \sin \omega t$$

$$i = I_m \sin \omega t$$

Lagging

In the figure shown, the zero point of the current waveform is after the zero point of the voltage waveform. Hence the current is lagging behind the voltage. The waveform, phasor and equation representation is as shown.

$$v = V_m \sin \omega t$$

$$i = I_m \sin (\omega t - \phi)$$

Lagging

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In the figure shown, the zero point of the current waveform is after the zero point of the voltage waveform. Hence the current is lagging behind the voltage. The waveform, phasor and equation representation is as shown.

$$v = V_m \sin \omega t$$

$$i = I_m \sin (\omega t + \phi)$$

4.5 PHASOR REPRESENTATION OF ALTERNATING QUANTITIES

A phasor is a rotating vector. Apart from the fact that a phasor rotates at a constant velocity, it has exactly the same properties as any other vector. Thus its length corresponds to the magnitude of a quantity. It has one end arrowed, to show the direction of action of the quantity.

Consider two such rotating vectors, v_1 and v_2 , rotating at the same angular velocity, rad/s. Let them rotate in a counterclockwise direction, with v_2 lagging behind v_1 by $\pi/6$ radian (30°).

This situation is illustrated in Fig.7 .

Figure 7

Notes

- Any a.c. quantity can be represented by a phasor, provided that it is a sine wave.

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- Any number of a.c. voltages and/or currents may be shown on the same phasor diagram, provided that they are all of the same frequency.
- Figure 7 shows a counterclockwise arrow, with ω rad/s. This has been shown here to emphasise the point that phasors must rotate in this direction only. It is normal practice to omit this from the diagram.
- When dealing with a.c. circuits, r.m.s. values are used almost exclusively. In this case, it is normal to draw the phasors to lengths that correspond to r.m.s. values.

Example 4: Four currents are as shown below. Draw to scale the corresponding phasor diagram.

$$i_1 = 2.5 \sin(\omega t + \pi/4) \text{ amp}, i_2 = 4 \sin(\omega t - \pi/3) \text{ amp}$$

$$i_3 = 6 \sin \omega t \text{ amp}, i_4 = 3 \cos \omega t \text{ amp}$$

Addition of Alternating Quantities

Consider two alternating currents, $i_1 = I_{m1} \sin \omega t$ amp and $i_2 = I_{m2} \sin (\omega t + \pi/4)$ amp, that are to be added together. There are three methods of doing this, as listed below.

> Plotting them on graph paper. Their ordinates are then added together, and the resultant waveform plotted. This is illustrated in Fig. 8. The amplitude, I_m , and the phase angle, θ , of the resultant current are then measured from the two axes.

> Drawing a scaled phasor diagram, as illustrated in Fig.9. The resultant is found by completing the parallelogram of vectors. The amplitude and phase angle are then measured on the diagram.

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> Resolving the two currents, into horizontal and vertical components, and applying Pythagoras theorem. This method involves using a sketch of the phasor diagram, followed by a purely mathematical process. This phasor diagram, including the identification of the horizontal and vertical components, is shown in Fig. 9.

The triangle of H.C., V.C., and the resultant current, is shown in Fig.10. From this, we can apply Pythagoras theorem to determine the amplitude and phase angle, thus

Figure 10

The graphical technique is very time-consuming (even for the addition of only two quantities). The accuracy also leaves much to be desired, a phasor diagram, drawn to scale, can be the quickest method of solution. However, it does require considerable care, in order to ensure a reasonable degree of accuracy, The use of the resolution of phasors is, with practice, a rapid

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technique, and yields a high degree of accuracy. Unless specified otherwise it is the technique you should use.

Example 5: Determine the phasor sum of the two voltages specified below.

$$V_1 = 25 \sin(\omega t + \pi/2)$$

$$V_2 = 15 \sin(\omega t - \pi/6)$$

Example 6: Calculate the phasor sum of the three currents listed below.

$$i_1 = 6 \sin \omega t \text{ A}$$

$$i_2 = 8 \sin(\omega t - \pi/2) \text{ A}$$

$$i_3 = 4 \sin(\omega t + \pi/6) \text{ A}$$

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4.6 A.C. THROUGH RESISTANCE, INDUCTANCE AND CAPACITANCE

A.C. Through Pure Ohmic Resistance Alone

The circuit is shown in Fig. 13. Let the applied voltage be given by the equation.

$$v = V_m \sin \omega t$$

Let R = ohmic resistance ; i = instantaneous current Obviously, the applied voltage has to supply ohmic voltage drop only. Hence for equilibrium $v = iR$;

$$v = V_m \sin \omega t = iR$$

$$i =$$

$$i = \frac{V_m}{R} \sin \omega t$$

$$i =$$

$$i =$$

$$i =$$

$$i =$$

Hence,

$$i = \frac{V_m}{R} \sin \omega t$$

Instantaneous power, $p = vi = \frac{V_m}{R} \sin^2 \omega t$

$$p =$$

$$p = \frac{V_m^2}{2R} (1 - \cos 2\omega t)$$

$$p =$$

Average value of $\cos 2\omega t$ is zero

$$P = \frac{1}{T} \int_0^T p dt$$

$$P = \frac{1}{T} \int_0^T \frac{V_m^2}{2R} (1 - \cos 2\omega t) dt$$

$$P = \frac{V_m^2}{2R}$$

Where V = r.m.s. value of applied voltage.

$I = \text{r.m.s. value of the current.}$

Example 6: A 60-Hz voltage of 115 V (r.m.s.) is impressed on a 100 ohm resistance : (i) Write the time equations for the voltage and the resulting current. Let the zero point of the voltage

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wave be at $t = 0$ (ii) Show the voltage and current on a time diagram. (iii) Show the voltage and current on a phasor diagram

A.C. Through Pure Inductance Alone

Whenever an alternating voltage is applied to a purely inductive coil, a back e.m.f. is produced due to the self-inductance of the coil. The back e.m.f., at every step, opposes the rise or fall of current through the coil. As there is no ohmic voltage drop, the applied voltage has to overcome this self-induced e.m.f. only. So at every step

$V =$

$=$

$V =$

$V =$

$V =$

$V =$

$V =$

$V =$

$=$

$V =$

$V =$

$V = \int$

$V =$

$V =$

$V = (-\sin(\omega t - 2)) = \sin(\omega t - 2)$

$V =$

$V =$

$V = \sin(\omega t - 2)$

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The maximum value of the instantaneous power is $V_m I_m / 2$

A.C. Through Pure Capacitance Alone

When an alternating voltage is applied to the plates of a capacitor, the capacitor is charged first in one direction and then in the opposite direction.

$V =$

$V =$

$i = I_m \sin(\omega t + \phi)$

$i = I_m \cos \phi$

$i = I_m \sin \phi$

$i = I_m \cos \phi$

$i = I_m \sin(\omega t + \phi)$

$i = I_m \cos \phi$

$i = I_m \sin \phi$

$i = I_m \cos \phi$

$i = I_m \sin(\omega t + \phi)$

$i = I_m \cos \phi$

The denominator $X_C = 1/\omega C$ is known as capacitive reactance and is in ohms.

The maximum value of the instantaneous power is $V_m I_m / 2$.

Example 7: A 50-Hz voltage of 230 volts effective value is impressed on a capacitance of 26.5 μF . (a) Write the time equations for the voltage and the resulting current. Let the zero axis of the voltage wave be at $t = 0$. (b) Show the voltage and current on a time diagram. (c) Show the voltage and current on a phasor diagram.

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A.C. Through Resistance and Inductance

A pure resistance R and a pure inductive coil of inductance L are shown connected in series in Fig. 14

Figure 14

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{(R)^2 + (\omega L)^2} = \sqrt{R^2 + \omega^2 L^2}$$

$Z = \sqrt{R^2 + \omega^2 L^2}$

$Z = \sqrt{R^2 + \omega^2 L^2}$

$Z = \sqrt{R^2 + \omega^2 L^2}$

The quantity $\sqrt{R^2 + \omega^2 L^2}$ is known as the impedance (Z) of the circuit.

$\tan \phi = \frac{X_L}{R}$

$\tan \phi = \frac{X_L}{R}$

$\tan \phi = \frac{X_L}{R}$

The mean power consumed by the circuit is given by the product of V and that component of the current I which is in phase with V .

So

$$P = V \times I \cos \phi$$

The term ' $\cos \phi$ ' is called the power factor of the circuit.

Active, Reactive and Apparent Power

Let a series R-L circuit draw a current of I when an alternating voltage of r.m.s. value V is applied to it. Suppose that current lags behind the applied voltage by ϕ . The three powers

drawn by the circuit are as under:

i. Apparent power (S) It is given by the product of r.m.s. values of applied voltage and circuit current. $\therefore S = VI = (IZ) \cdot I = I^2 Z$ volt-amperes (VA)

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ii. Active power (P or W) It is the power which is actually dissipated in the circuit resistance. $P = I^2 R = VI \cos \phi$ watts

iii. Reactive power (Q) It is the power developed in the inductive reactance of the circuit.

$Q = I^2 X_L = I^2 Z \sin \phi = I \cdot (IZ) \sin \phi = VI \sin \phi$ volt-amperes-reactive (VAR)

These three powers are shown in the power triangle of Fig 1 from where it can be seen that $S^2 = P^2 + Q^2$.

Q-factor of a Coil

Reciprocal of power factor is called the Q-factor of a coil or its figure of merit. It is also known as quality factor of the coil.

Q factor = $1/\text{power factor}$ OR $1/\cos \phi$

Also $Q = 2\pi (\text{maximum energy stored})/(\text{energy dissipated per cycle})$

Example 8: In a series circuit containing pure resistance and a pure inductance, the current and the voltage are expressed as : $i(t) = 5 \sin(314t + 2\pi/3)$ and $v(t) = 15 \sin(314t + 5\pi/6)$ (a) What is the impedance of the circuit ? (b) What is the value of the resistance? (c) What is the inductance in henrys? (d) What is the average power drawn by the circuit? (e) What is the power factor?

R-C Series circuit

Consider an AC circuit with a resistance R and a capacitance C connected in series as shown in the figure. The alternating voltage v is given by

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$$v = V_m \sin \omega t$$

The current flowing in the circuit is i. The voltage across the resistor is V_R and that across the capacitor is V_C .

$V_R = IR$ is in phase with I

$V_C = IX_C$ lags behind the current by 90 degrees

With the above information, the phasor diagram can be drawn as shown.

The current I is taken as the reference phasor. The voltage V_R is in phase with I and the voltage V_C lags behind the current by 90° . The resultant voltage V can be drawn as shown in the figure. From the phasor diagram we observe that the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ .

The waveform and equations for an RC series circuit can be drawn as below.

$$V = V_R \sin \omega t$$

$$V = V_R \sin \omega t + V_C$$

From the phasor diagram, the expressions for the resultant voltage V and the angle can be derived as follows.

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$$V = \sqrt{V_R^2 + V_C^2}$$

$$V_R = V \cos \phi, V_C = V \sin \phi$$

$$V = \sqrt{V_R^2 + V_C^2}$$

$$V = \sqrt{I^2 R^2 + I^2 X_C^2}$$

$$V = I Z$$

Where the impedance Z is

$$Z = \sqrt{R^2 + X_C^2}$$

The phase angle

Hence the power in an RC series circuit is consumed only in the resistance. The capacitance does not consume any power.

Impedance Triangle

We can derive a triangle called the impedance triangle from the phasor diagram of an RC series circuit as shown

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Example 9: A Capacitor of capacitance $79.5\mu\text{F}$ is connected in series with a non inductive resistance of $30\ \Omega$ across a 100V , 50Hz supply. Find (i) impedance (ii) current (iii) phase angle (iv) Equation for the instantaneous value of current.

R-L-C Series circuit

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Consider an AC circuit with a resistance R , an inductance L and a capacitance C connected in series as shown in the figure. The alternating voltage v is given by

The current flowing in the circuit is i . The voltage across the resistor is V_R , the voltage across the inductor is V_L and that across the capacitor is V_C .

$V_R = IR$ is in phase with I

$V_L = I X_L$ leads the current by 90 degrees

$V_C = I X_C$ lags behind the current by 90 degrees

With the above information, the phasor diagram can be drawn as shown. The current I is taken as the reference phasor. The voltage V_R is in phase with I , the voltage V_L leads the current by

90° and the voltage V_C lags behind the current by 90° . There are two cases that can occur $V_L > V_C$ and $V_L < V_C$ depending on the values of X_L and X_C . And hence there are two possible phasor diagrams. The phasor $V_L - V_C$ or $V_C - V_L$ is drawn and then the resultant voltage V is drawn

From the phasor diagram we observe that when $V_L > V_C$, the voltage leads the current by an angle Φ or in other words the current lags behind the voltage by an angle Φ . When $V_L < V_C$, the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ .

From the phasor diagram, the expressions for the resultant voltage V and the angle Φ can be derived as follows.

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From the expression for phase angle, we can derive the following three cases

Case (i): When $X_L > X_C$

The phase angle Φ is positive and the circuit is inductive. The circuit behaves like a series RL circuit.

Case (ii): When $X_L < X_C$

The phase angle Φ is negative and the circuit is capacitive. The circuit behaves like a series RC circuit.

Case (iii): When $X_L = X_C$

The phase angle $\Phi = 0$ and the circuit is purely resistive. The circuit behaves like a pure resistive circuit.

The voltage and the current can be represented by the following equations. The angle Φ is positive or negative depending on the circuit elements.

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Average power

Hence the power in an RLC series circuit is consumed only in the resistance. The inductance and the capacitance do not consume any power.

Phasor algebra for RLC series circuit

Example 10: A 230 V, 50 Hz ac supply is applied to a coil of 0.06 H inductance and 2.5 resistance connected in series with a 6.8 μF capacitor. Calculate (i) Impedance (ii) Current (iii) Phase angle between current and voltage (iv) power factor (v) power consumed

Example 11: A resistance R , an inductance $L = 0.01$ H and a capacitance C are connected in series. When an alternating voltage $v = 400\sin(3000t - 20^\circ)$ is applied to the series combination, the current flowing is $10\sqrt{2}\sin(3000t - 65^\circ)$. Find the values of R and C .

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Example 12: A coil of pf 0.6 is in series with a $100\mu\text{F}$ capacitor. When connected to a 50Hz supply, the potential difference across the coil is equal to the potential difference across the capacitor. Find the resistance and inductance of the coil.

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